## Investigation: COMPANHIA ENERGÉTICA DE BRASÍLIA - CEB (CVM code 14451), fiscal year 2010, as it was on 2012-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Energia Elétrica (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2010 DFP, version 2, filed on 2011-07-18; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2011-07-15); 2 (2011-07-18).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=10177

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DEPI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: 1.076 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 77 (n = 51)
- Percentile in group and year: 68 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.016 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 40 (n = 51)
- Percentile in group and year: 66 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.966 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 30 (n = 52)
- Percentile in group and year: 42 (n = 471)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.125 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 74 (n = 51)

Percentile in group and year: 47 (n = 468)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: missing account: depreciation, cash flow lines in section 6.01.01
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: 1.026 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 54 (n = 51)
Percentile in group and year: 63 (n = 465)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.044 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 51 (n = 54)
Percentile in group and year: 42 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.005 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 54 (n = 53)
Percentile in group and year: 42 (n = 481)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 2.4 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 18.4 days)
Percentile in sector and year: 56 (n = 51)
Percentile in group and year: 60 (n = 439)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 9.10 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 79 (n = 51)
Percentile in group and year: 58 (n = 435)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.045 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 51 (n = 54)
Percentile in group and year: 43 (n = 487)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 0.167 (index)
Components: X1 = −0.084; X2 = −0.025; X3 = 0.039; X4 = 0.508 (indices)
Percentile in sector and year: 8 (n = 56)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (5 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA: —
- Cash above earnings (ACCRUAL): 1
- Change in leverage: —
- Change in current ratio: 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: —
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: 2 (threshold of R$21.2 million, 1% of total assets of the version used). Values in R$ million:
DRE 3.01 Receita de Venda de Bens e/ou Serviços: 1,839.3 in the version used (v2, filed on 2011-07-18); 1,284.4 in the next DFP's comparative (v2, filed on 2012-04-09); difference of −554.9. Different value in the following year's comparative (restatement or reclassification).
DFC (direct method) 6.01 Caixa Líquido Atividades Operacionais: 113.2 in the version used (v2, filed on 2011-07-18); 86.4 in the next DFP's comparative (v2, filed on 2012-04-09); difference of −26.8. Different value in the following year's comparative (restatement or reclassification).
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller

differences) stay in the full list.
Full list: 54 accounts above the R$1,000 tolerance (group a, totals and key lines: 6; group b, reclassifications: 48), in the annex at the end.

## What was known on the date

- FCA 2010, v1: 2010-03-22
- FCA 2010, v2: 2010-08-19
- DFP 2010, v1: 2011-07-15
- DFP 2010, v2: 2011-07-18
- DFP 2011, v1: 2012-03-31
- DFP 2011, v2: 2012-04-09

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).
Excerpt checked literally: "Em nossa opinião, tais ajustes são apropriados e foram adequadamente efetuados."
Auditor in the 2010 FCA: KPMG Auditores Independentes.
Auditor in the 2011 FCA: KPMG Auditores Independentes.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 20 held-out cases, a single measurement with the frozen reader (c99c953): precision 0.962, coverage 0.962, literal check on 132 of 132 excerpts, no natural-person name (docs/leitor_medida.json).

Explanatory notes · Related parties
Note: 27 TRANSAÇÕES COM PARTES RELACIONADAS (pages 69 to 71)

- [parent company] “Controladora final” (p. 69, original excerpt in Portuguese)
- Excerpt omitted; see page 70 of the document (4 excerpts)
- [key management personnel] “Operações com pessoal-chave da Administração” (p. 70, original excerpt in Portuguese)
- [key management personnel; other related parties] “A Companhia não possui transações de empréstimos ou outras transações com diretores/conselheiros ou familiares imediatos.” (p. 70, original excerpt in Portuguese)
- [key management personnel] “Conselho de Administração 237 275 345” (p. 70, original excerpt in Portuguese)
- [key management personnel] “Conselho Fiscal 133 145 148” (p. 70, original excerpt in Portuguese)
- [key management personnel] “Diretoria 579285310 Total 949705803” (p. 70, original excerpt in Portuguese)
- [key management personnel] “A Companhia não concede benefícios pós-emprego aos administradores

e conselheiros.” (p. 70, original excerpt in Portuguese)
- [subsidiaries; associates] “A relação de suas participações em empresas coligadas e controladas está demonstrada na Nota Explicativa 01.” (p. 70, original excerpt in Portuguese)
- [other related parties] “Distribuição com a CEB?Holding controle comum)” (p. 70, original excerpt in Portuguese)
- Excerpt omitted; see page 71 of the document (4 excerpts)
- [other related parties] “a CEB Holding controle comum)” (p. 71, original excerpt in Portuguese)
- [other related parties] “(c) controle comum) 35 36 ?” (p. 71, original excerpt in Portuguese)
- [subsidiaries] “(b) Despesas de Pessoal – Referem-se ao ressarcimento dos custos relativos ao pessoal cedido pela Controlada CEB Distribuição S.A..” (p. 71, original excerpt in Portuguese)

Fields of item 18 of CPC 05 (R1)
- Excerpt omitted; see page 70 of the document (6 excerpts)
- [balances] “Os saldos das operações ativas e passivas da CEB com as partes relacionadas no período são os seguintes:” (p. 70, original excerpt in Portuguese)
- Excerpt omitted; see page 71 of the document (7 excerpts)
- [amount of the transactions] “(b) Despesas de Pessoal – Referem-se ao ressarcimento dos custos relativos ao pessoal cedido pela Controlada CEB Distribuição S.A..” (p. 71, original excerpt in Portuguese)

Market terms
- “Essas transações com partes relacionadas são praticadas em condições normais de mercado, vigentes nas respectivas datas e não envolvem riscos anormais de recebimento.” (p. 70, original excerpt in Portuguese)

The sentence about market terms is recorded as a fact, without proof (CPC 05 (R1), item 23).

Explanatory notes · Reverse factoring and forfaiting

The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report

Type of opinion: qualified
- [basis for the opinion] “Base para opinião com ressalva sobre as demonstrações financeiras individuais e consolidadas 5. Parcela do saldo consolidado da rubrica “Depósitos e bloqueios judiciais - Ativo não circulante”, no valor de R$ 7,4 milhões, em 31 de dezembro de 2010 (Nota Explicativa nº 13), está em processo de conciliação com os respectivos documentos comprobatórios. O atual estágio do processo de conciliação não permitiu a aplicação de procedimentos de auditoria para avaliar a adequação do referido saldo. 6. Os montantes de ICMS pagos na aquisição de bens utilizados na atividade de distribuição de energia da controlada CEB Distribuição S.A. são passíveis de ser compensados com os débitos do ICMS sobre faturamento, nos termos e critérios estabelecidos pela legislação fiscal vigente. Neste contexto, a Companhia vem compensano, mensalmente, parte do referido crédito e mantém registrados no ativo valores avaliados pela Administração como possíveis de compensação futura (Nota Explicativa nº 10).” (cut at the end of a sentence, original excerpt in Portuguese)
- Excerpt omitted; see the auditor's report in the document
- [other matters] “Outros assuntos Demonstrações do valor adicionado 17. Examinamos, também, as demonstrações individuais e consolidadas do valor adicionado (DVA) referentes ao exercício findo em 31 de dezembro de 2010, cuja apresentação é requerida pela legislação societária brasileira para companhias de capital aberto e como informação suplementar pelas IFRS, que não requerem a apresentação da DVA. Essas demonstrações foram submetidas aos mesmos procedimentos de auditoria descritos anteriormente e, em nossa opinião, estão adequadamente apresentadas, em todos os aspectos relevantes em relação às demonstrações financeiras tomadas em conjunto. Auditoria dos valores correspondentes aos exercícios anteriores 18. As demonstrações financeiras, individuais e consolidadas, correspondentes ao exercício findo em 31 de dezembro de 2009, que foram ajustadas às novas práticas contábeis, conforme demonstrado na Nota Explicativa n° 3, foram anteriormente por nós auditadas de acordo com as normas de auditoria vigentes quando da emissão do relatório em 30 de março de 2010.” (cut at the end of a

sentence, original excerpt in Portuguese)

Reference Form
The Reference Form for this fiscal year was not available on the date. It is the only version available in the open data; earlier versions are not there.

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, header lines (years, column titles) may come out as excerpts; the held-out cases had no tables with multi-line headers.

## How the engine was tested
In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits
- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations
- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

## Engine
motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-1 (commit c99c953). Generated at 2026-11-12T03:37:14+00:00.

## Today's information (outside the reference date)
Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: COMPANHIA ENERGÉTICA DE BRASÍLIA - CEB.

## Annex: full list of differences in the following year's comparative (R$ million)
(a) DRE 3.01 Receita de Venda de Bens e/ou Serviços: 1,839.3 in the version used, 1,284.4 in the comparative (difference of −554.9)
(a) DFC (direct method) 6.01 Caixa Líquido Atividades Operacionais: 113.2 in the version used, 86.4 in the comparative (difference of −26.8)
(a) BPP 2.03 Patrimônio Líquido Consolidado: 714.3 in the version used, 724.6 in the comparative (difference of 10.3)
(a) BPA 1 Ativo Total: 2,119.9 in the version used, 2,130.2 in the comparative (difference of 10.3)
(a) DRE 3.11 Lucro/Prejuízo Consolidado do Período: 19.6 in the version used, 18.8 in the comparative (difference of −0.8)

(a) BPP 2.01 Passivo Circulante: 627.9 in the version used, 627.9 in the comparative (difference of −0.0)
(b) DRE 3.04.02 Despesas Gerais e Administrativas: −1,057.6 in the version used, −129.3 in the comparative (difference of 928.3)
(b) DRE 3.04 Despesas/Receitas Operacionais: −1,112.3 in the version used, −211.2 in the comparative (difference of 901.1)
(b) DRE 3.03 Resultado Bruto: 1,196.0 in the version used, 295.1 in the comparative (difference of −900.9)
(b) DRE 3.02 Custo dos Bens e/ou Serviços Vendidos: −643.3 in the version used, −989.3 in the comparative (difference of −345.9)
(b) BPA 1.02.04.01.01 Contrato de Concessão: 386.1 in the version used, 244.7 in the comparative (difference of −141.4)
(b) DRE 3.04.05 Outras Despesas Operacionais: −0.0 in the version used, −82.8 in the comparative (difference of −82.8)
(b) DRE 3.04.04 Outras Receitas Operacionais: 0.3 in the version used, 55.0 in the comparative (difference of 54.7)
(b) BPA 1.02.02 Investimentos: 285.1 in the version used, 316.0 in the comparative (difference of 30.8)
(b) BPA 1.02.02.01 Participações Societárias: 285.1 in the version used, 316.0 in the comparative (difference of 30.8)
(b) DFC (direct method) 6.03 Caixa Líquido Atividades de Financiamento: −40.8 in the version used, −13.9 in the comparative (difference of 26.8)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 799.5 in the version used, 775.5 in the comparative (difference of −23.9)
(b) BPA 1.02.02.01.01 Participações em Coligadas: 283.4 in the version used, 302.6 in the comparative (difference of 19.2)
(b) DRE 3.02.01 Impostos e Contribuições: −643.3 in the version used, −661.1 in the comparative (difference of −17.7)
(b) BPA 1.02.01.03 Contas a Receber: 127.9 in the version used, 111.9 in the comparative (difference of −16.1)
(b) BPA 1.02.01.03.02 Outras Contas a Receber: 17.6 in the version used, 1.5 in the comparative (difference of −16.1)
(b) BPA 1.02.02.01.04 Outras Participações Societárias: 1.7 in the version used, 13.3 in the comparative (difference of 11.6)
(b) BPP 2 Passivo Total: 2,119.9 in the version used, 2,130.2 in the comparative (difference of 10.3)
(b) BPP 2.01.05.02.10 Outras Obrigações: 22.6 in the version used, 13.7 in the comparative (difference of −8.8)
(b) BPA 1.01.03 Contas a Receber: 321.2 in the version used, 329.1 in the comparative (difference of 7.9)
(b) BPA 1.01.03.01 Clientes: 321.2 in the version used, 329.1 in the comparative (difference of 7.9)
(b) BPA 1.01.03.01.09 Outros Créditos: 15.2 in the version used, 23.2 in the comparative (difference of 7.9)
(b) BPA 1.01 Ativo Circulante: 450.0 in the version used, 457.9 in the comparative (difference of 7.9)
(b) BPA 1.02.01.09 Outros Ativos Não Circulantes: 630.6 in the version used, 622.7 in the comparative (difference of −7.9)
(b) BPA 1.02.01.09.01 Ativos Não-Correntes a Venda: 286.9 in the version used, 279.0 in the comparative (difference of −7.9)
(b) BPP 2.03.09 Participação dos Acionistas Não Controladores: 224.3 in the version used, 231.6 in the comparative (difference of 7.3)
(b) BPA 1.02.03 Imobilizado: 197.4 in the version used, 192.9 in the comparative (difference of −4.5)
(b) BPA 1.02.03.01 Imobilizado em Operação: 197.4 in the version used, 192.9 in the comparative (difference of −4.5)

(b) BPP 2.01.05.02.01 Dividendos e JCP a Pagar: 5.3 in the version used, 9.7 in the comparative (difference of 4.4)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: −82.5 in the version used, −79.5 in the comparative (difference of 3.0)
(b) BPA 1.02 Ativo Não Circulante: 1,670.0 in the version used, 1,672.4 in the comparative (difference of 2.4)
(b) BPP 2.01.05 Outras Obrigações: 134.0 in the version used, 131.7 in the comparative (difference of −2.3)
(b) BPP 2.01.05.02 Outros: 134.0 in the version used, 131.7 in the comparative (difference of −2.3)
(b) BPP 2.01.04 Empréstimos e Financiamentos: 154.2 in the version used, 156.5 in the comparative (difference of 2.3)
(b) BPP 2.01.04.01 Empréstimos e Financiamentos: 154.2 in the version used, 156.5 in the comparative (difference of 2.3)
(b) BPP 2.01.04.01.01 Em Moeda Nacional: 154.2 in the version used, 156.5 in the comparative (difference of 2.3)
(b) BPP 2.01.05.02.05 Encargos Regulatórios: 106.2 in the version used, 108.3 in the comparative (difference of 2.1)
(b) BPA 1.02.04.01.02 Diferido: 1.9 in the version used, 0.0 in the comparative (difference of −1.9)
(b) DRE 3.06.01 Receitas Financeiras: 46.6 in the version used, 44.8 in the comparative (difference of −1.8)
(b) DRE 3.06 Resultado Financeiro: −43.6 in the version used, −44.6 in the comparative (difference of −1.0)
(b) DRE 3.04.06 Resultado de Equivalência Patrimonial: 7.9 in the version used, 8.9 in the comparative (difference of 0.9)
(b) DRE 3.07 Resultado Antes dos Tributos sobre o Lucro: 40.1 in the version used, 39.3 in the comparative (difference of −0.8)
(b) DRE 3.09 Resultado Líquido das Operações Continuadas: 19.6 in the version used, 18.8 in the comparative (difference of −0.8)
(b) DRE 3.06.02 Despesas Financeiras: −90.2 in the version used, −89.5 in the comparative (difference of 0.8)
(b) DRE 3.11.01 Atribuído a Sócios da Empresa Controladora: 2.8 in the version used, 2.0 in the comparative (difference of −0.7)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: 83.7 in the version used, 83.9 in the comparative (difference of 0.2)
(b) BPA 1.01.06 Tributos a Recuperar: 20.9 in the version used, 20.9 in the comparative (difference of −0.1)
(b) BPA 1.01.06.01 Tributos Correntes a Recuperar: 20.9 in the version used, 20.9 in the comparative (difference of −0.1)
(b) DRE 3.11.02 Atribuído a Sócios Não Controladores: 16.8 in the version used, 16.8 in the comparative (difference of −0.0)